Project Navi

Bachelor Designer & Developer

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Project Charter

Project Background and Context

Project Overview

The project aims to develop a mobile application designed to assist visually impaired users in their daily activities. The application seeks to improve autonomy, safety, and accessibility by providing voice-guided assistance, object recognition, text reading, and navigation support.

Identified Need

Visually impaired individuals often face challenges when navigating unfamiliar environments, identifying objects, reading printed information, and accessing essential services independently. While several accessibility applications already exist, most focus on specific functionalities and do not provide a complete assistance solution.

Target Users

The primary target audience consists of visually impaired and blind individuals seeking greater independence in their daily lives. Secondary users may include elderly people experiencing visual impairments and accessibility organizations interested in supporting inclusive technologies.

Opportunity Analysis Summary

The opportunity study highlighted a strong demand for accessible digital solutions. Benchmark analysis revealed that existing applications such as Seeing AI, Google Lookout, and Be My Eyes offer valuable assistance features but still present limitations in terms of real-time navigation, contextual awareness, or autonomy.

The SWOT analysis confirmed the relevance of the project by identifying significant strengths, including its social impact and accessibility-focused design. It also highlighted opportunities related to advances in artificial intelligence and increasing awareness of digital inclusion, while emphasizing the importance of reliability, data protection, and technical feasibility.

As a result, the project has been considered viable and relevant, justifying the transition to the project planning and definition phase.

Project Definition

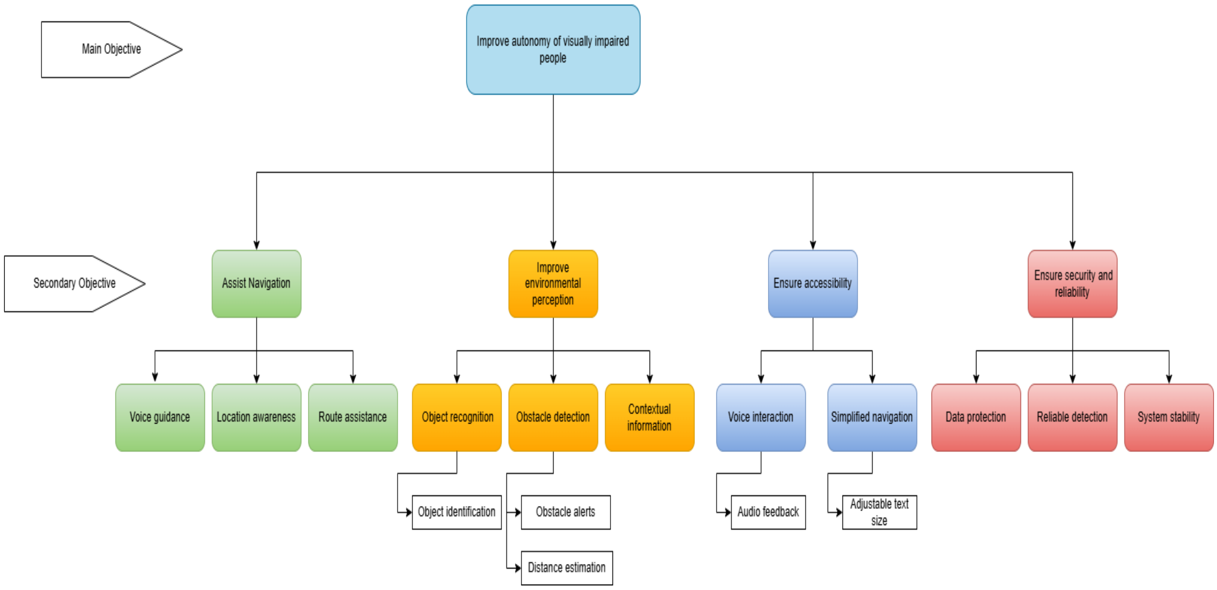
SMART Objectives

Objective	Specific	Measurable	Achievable	Relevant	Time
Develop an accessible mobile application for visually impaired users	AI-powered assistance and accessible interface	Functional prototype completed	Yes	Supports user autonomy	End of project
Provide real-time environmental information	Object and obstacle recognition	Detection accuracy tested and validated	Yes	Improves safety during mobility	

Facilitate independent navigation	Voice guidance and contextual notifications	Navigation scenarios successfully completed	Yes	Core project objective	
Ensure accessibility compliance	Accessible design principles implemented	Accessibility checklist validated	Yes	Essential for target users	Throughout development
Protect user data and privacy	Secure management of location and user data	Security requirements implemented	Yes	Regulatory and ethical requirement	Before deployment

The SMART objectives define the main goals of the project in a structured and measurable way. They ensure that each feature of the application contributes to improving accessibility and autonomy for visually impaired users. These objectives also help track progress throughout the development process.

Objective Tree



The objective tree provides a hierarchical representation of the project goals, from the main objective down to specific functionalities. It helps to visualize how each feature contributes to improving accessibility, navigation, and user autonomy.

QQOQCCP Analysis

The QQOQCCP analysis helps to clearly define the context of the project by identifying the target users, needs, environment, and purpose of the application. It ensures a better understanding of

the problem and confirms the relevance of the proposed solution for visually impaired users.

Question	**Answer**
Who ?	Visually impaired and partially sighted individuals seeking

	greater independence in daily mobility.
What ?	An AI-powered mobile application providing navigation assistance and environmental awareness.
Where ?	In public spaces, urban environments, transport areas, and everyday locations.
When ?	During daily travel and navigation activities.
How ?	Through smartphone camera, artificial intelligence, voice feedback, and location services.
How much ?	Initial deployment targets a limited user base for testing before broader adoption.
Why ?	To increase autonomy, safety, accessibility, and social inclusion for visually impaired users.

Functional Scope Definition

Ref	Type	Name	Description	Complexity	Priority	Ver.
F1	Function	Voice guidance	Real-time spoken navigation assistance	High	High	V1
F2	Function	Object Recognition	AI-based identification of nearby objects	High	High	V1
F3		Obstacle Detection	Detection of obstacles and warning notifications	High	High	V1
F4	Screen	Website	App. website	Low	Medium	V1
F5		Offline Assistance	Provide limited functionality without internet connection	High	Low	V2

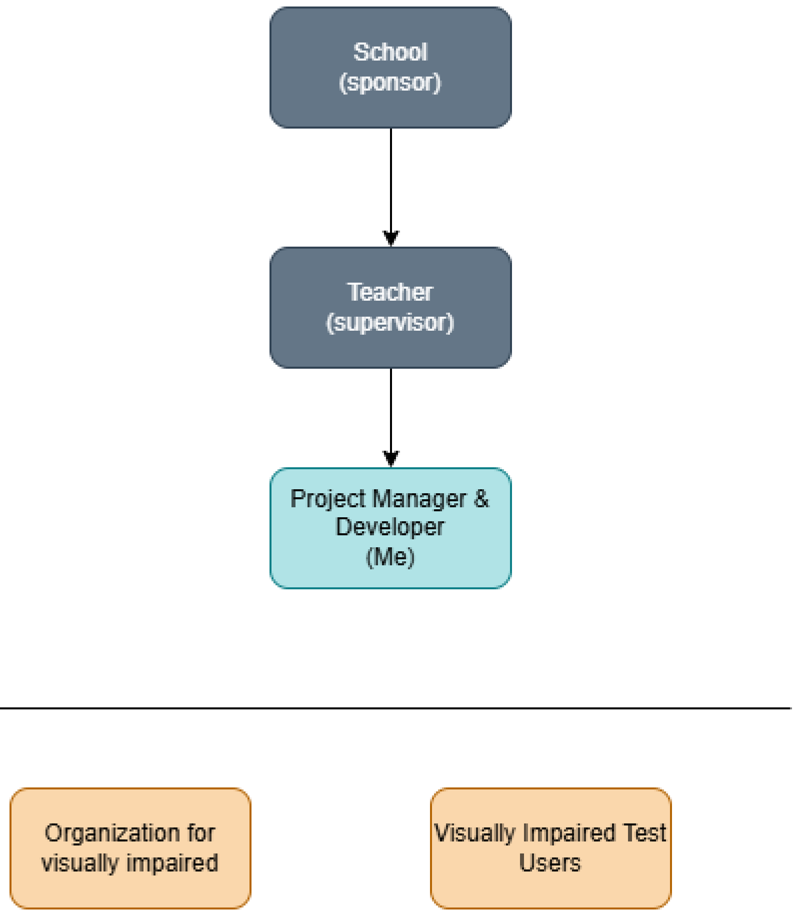
The functional scope defines all features included in the first version (V1) of the application. It focuses on essential functionalities required to support visually impaired users, such as accessibility features, navigation assistance, and AI-based environment analysis.

Each feature is prioritized and classified according to its importance and complexity in order to ensure a progressive development approach. Additional features may be considered in future versions (V2), depending on technical feasibility and project constraints.

Non-essential functionalities, such as social features or advanced integrations, are excluded from the current scope to maintain focus on the core objectives of accessibility and user autonomy

Project Organization

Team Structure



The project is developed by a single student acting as Project Manager and Developer, responsible for planning, design, implementation, testing, and documentation activities. The project is supervised by the academic supervisor, who provides guidance, monitors progress, and validates key deliverables throughout the project lifecycle.

Although the development is carried out individually, several stakeholders may contribute valuable feedback. Accessibility associations can provide recommendations regarding accessibility requirements, while visually impaired users may participate in testing activities to evaluate usability and identify potential improvements. Their feedback helps ensure that the application addresses real user needs and supports digital inclusion.

Planning / Roadmap (tableau ou gantt)

Phase	**Activities**	**Duration**
Analysis	Benchmark of existing solutions, SWOT analysis, definition of user needs	Week 1-2
Project definition	SMART objectives, functional scope, planning setup	Week 3
Design	UI/UX wireframes, application structure, accessibility design	Week 4-5
Development	Mobile application development (voice guidance, GPS, AI features)	Week 6-10
Testing	Functional testing, accessibility testing with users	Week 11
Finalization	Bug fixes, documentation, final report	Week 12

This planning outlines the main phases of the project over an estimated 12-week period. It starts with analysis and requirement gathering, followed by design and system architecture. The development phase represents the core of the project, where main functionalities such as voice guidance, GPS navigation, and AI-based assistance are implemented. Testing is then carried out to ensure functionality and accessibility compliance. Finally, the project is completed with documentation and final adjustments before delivery.

Budget Estimation

Category	**Estimated Cost (€)**
Hosting / deployment	50
API services (AI / recognition)	100
Development tools	0
Testing devices	0
Miscellaneous	50
Total	200

The budget presented above is an estimated cost for the development and deployment of the application. Since this is an academic project, most tools and development resources are free or provided by educational institutions. Costs mainly concern optional external services such as AI APIs or hosting infrastructure.

Conclusion

This project definition confirms the feasibility and relevance of developing an accessible mobile application for visually impaired users. The defined objectives, functional scope, and planning provide a clear framework for the development process.

The next phase will focus on the detailed design and implementation of the application, ensuring that accessibility and user autonomy remain at the center of the development approach.